Lecture 7. Regularization

Variable selection, ridge, lasso, elastic net, choosing the penalty

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7.1 Beyond least squares

Motivation. With many predictors, selecting or shrinking coefficients beats plain least squares.

The linear model $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$ is often competitive; other fitting procedures can improve accuracy and interpretability:

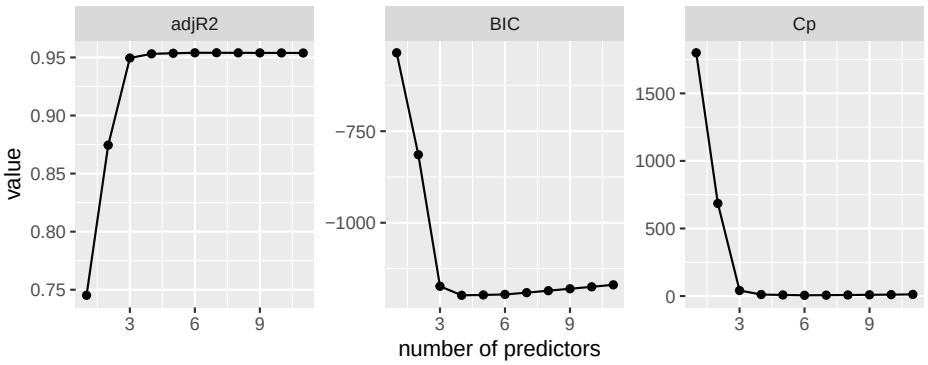
1. **subset selection**: use only some predictors;
2. **shrinkage** (regularization): shrink coefficients toward zero; some forms also select;
3. dimension reduction (not covered).

7.2 Subset selection

Fit is measured by RSS $= \sum_i (y_i - \hat{y}_i)^2$.

Best subset selection fits all 2^p models: null M_0; for each k the best M_k (smallest RSS) of the $\binom{p}{k}$ models; then chooses among $M_0, \ldots, M_p$ by an estimate of *test* error (validation, C_p, BIC, adjusted R^2), since RSS always picks the full model. `ISLR2::Credit` (balances of 400 people), `leaps::regsubsets()`:

```
fit_full <- regsubsets(Balance ~ ., data = Credit, nvmax = 11)
s <- summary(fit_full)
tibble(d = 1:11, Cp = s$cp, BIC = s$bic, adjR2 = s$adjr2) %>%
  pivot_longer(-d) %>%
  ggplot(aes(d, value)) + geom_line() + geom_point() +
  facet_wrap(~ name, scales = "free_y") + labs(x = "number of predictors")
```




```
c(Cp = which.min(s$cp), BIC = which.min(s$bic), adjR2 = which.max(s$adjr2))
```

```
##    Cp   BIC adjR2
##     6     4     7
```

BIC picks the smallest; all flatten after about four predictors.

Stepwise selection. Best subset is infeasible for large p and its search invites overfitting. **Forward stepwise** adds the best variable one at a time: $1 + \sum_{k=0}^{p-1}(p-k) = 1 + p(p+1)/2$ models (211 instead of 2^{20} for $p = 20$), not guaranteed optimal. **Backward stepwise** removes the least useful variable (needs $n > p$). **Hybrid** methods alternate adding and removing.

```
fwd <- regsubsets(Balance ~ ., data = Credit, nvmax = 11, method = "forward")
names(coef(fit_full, 4)); names(coef(fwd, 4))
```

```
## [1] "(Intercept)" "Income"      "Limit"       "Cards"       "StudentYes"
```

```
## [1] "(Intercept)" "Income"      "Limit"       "Rating"      "StudentYes"
```

Forward stepwise keeps `Rating` (entered first) instead of `Cards`.

7.2.1 Choosing the optimal model

Test error is estimated *indirectly* (adjusting training error) or *directly* (validation, CV). With d predictors and $\hat{\sigma}^2$ from the full model:

$$C_p = \frac{1}{n}(\mathrm{RSS} + 2d\hat{\sigma}^2), \qquad \mathrm{BIC} = \frac{1}{n}(\mathrm{RSS} + \log(n)\, d\hat{\sigma}^2), \qquad R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{RSS}/(n-d-1)}{\mathrm{TSS}/(n-1)}.$$

- C_p penalises the optimistic training RSS by $2d\hat{\sigma}^2$; AIC (any ML model) is proportional to it for Gaussian linear models.
- BIC uses $\log n > 2$ ($n > 7$): smaller models.
- Adjusted R^2 is *maximised*, i.e. $\mathrm{RSS}/(n-d-1)$ minimised.

Why $2d\hat{\sigma}^2$ [Supplement]

For least squares with d predictors (plus intercept), $\mathbb{E}[\mathrm{RSS}] = (n-d-1)\sigma^2$, while the expected squared error on new responses at the same x's is $(n+d+1)\sigma^2$. The optimism $2(d+1)\sigma^2$ is what C_p adds back (up to the constant $2\sigma^2$; Exercise 7.3).

Cross-validation needs neither σ^2 nor a parameter count.

7.3 Regularization

Motivation. Regularization replaces yes–no variable choices by a continuous penalty.

Regularization constrains $f_\beta(x)$ against overfitting. Counting nonzero coefficients,

$$\hat{\beta} = \arg\min_\beta \frac{1}{n}\sum_i L(y_i, f_\beta(x_i)) \quad \text{subject to} \quad \sum_{j=1}^{p} \mathbb{1}(\beta_j \neq 0) \leq c,$$

is best subset selection, NP-hard with a non-convex L_0 constraint. Convex substitutes:

- $\sum_j |\beta_j| \leq c_1$ (**lasso**): sparse;
- $\sum_j \beta_j^2 \leq c_2$ (**ridge**): no zeros;
- the **elastic net**: a compromise.

The lasso combines L_2 *loss* with L_1 *penalty*.

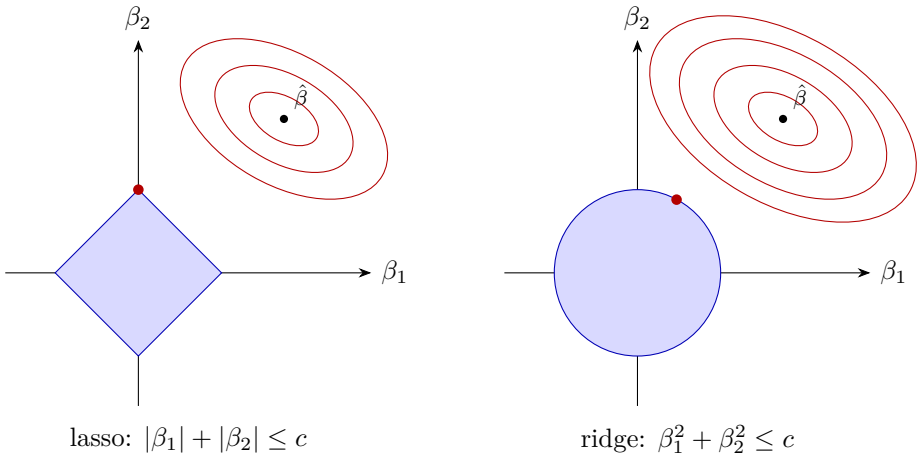


Figure. *RSS contours around $\hat{\beta}$ first touch the lasso diamond at a corner ($\beta_1 = 0$), the ridge disc at a generic point.* [Supplement]

7.3.1 Penalized form

With complexity $R(\beta)$, the constrained problem is equivalent to

$$\hat{\beta} = \arg\min_{\beta} \Big[\frac{1}{n}\sum_{i=1}^{n} L(y_i, \hat{y}_i) + \lambda R(\beta)\Big],$$

with **tuning parameter** $\lambda \ge 0$: ridge $\lambda \sum_j \beta_j^2$, lasso $\lambda \sum_j |\beta_j|$ (intercept unpenalised, predictors standardised). The penalty **shrinks** estimates: $\lambda = 0$ is least squares, $\lambda \to \infty$ gives zero. λ is chosen by cross-validation; shrinkage trades bias for variance and can beat least squares with many predictors.

Proposition 7.1 (orthonormal case)

Let $n = p$, $X = I$ and no intercept, so least squares minimises $\sum_j (y_j - \beta_j)^2$ with solution $\hat{\beta}_j = y_j$. Then ridge and lasso give

$$\hat{\beta}_j^{\text{ridge}} = \frac{y_j}{1+\lambda}, \qquad \hat{\beta}_j^{\text{lasso}} = \begin{cases} y_j - \lambda/2, & y_j > \lambda/2, \\ y_j + \lambda/2, & y_j < -\lambda/2, \\ 0, & |y_j| \le \lambda/2. \end{cases}$$

Proof [Supplement]**.** Both objectives separate by coordinate. Ridge: $\frac{d}{d\beta}\{(y-\beta)^2 + \lambda\beta^2\} = -2(y-\beta) + 2\lambda\beta = 0$ gives $\beta = y/(1+\lambda)$. Lasso: $h(\beta) = (y-\beta)^2 + \lambda|\beta|$ is convex. For $\beta > 0$, $h'(\beta) = -2(y-\beta) + \lambda$ vanishes at $\beta = y - \lambda/2$, admissible iff $y > \lambda/2$; symmetrically for $\beta < 0$. If $|y| \le \lambda/2$, the one-sided derivatives at 0 are $-2y + \lambda \ge 0$ (right) and $-2y - \lambda \le 0$ (left), so 0 is the minimum. ■

Ridge shrinks proportionally; the lasso shifts by $\lambda/2$ and zeroes small ones (**soft thresholding**), hence selection.

7.4 Example: delays of flights to California

Arrival delays from EWR/JFK to four California airports on hour, origin, destination, carrier, month, weekday (slides: `tidymodels`; here `lm()`, `glmnet()`).

```
California <- flights %>%
  filter(dest %in% c("LAX", "SFO", "OAK", "SJC"), !is.na(arr_delay)) %>%
  mutate(day = as.Date(time_hour),
         dow = as.character(lubridate::wday(day, label = TRUE)),
         month = as.factor(month), hour = as.factor(hour))
dim(California)
```

```
## [1] 29836    20
```

```
set.seed(386)
idx <- sample(nrow(California), size = round(0.7 * nrow(California)))
train <- California[idx, ]; test <- California[-idx, ]
flight_model <- arr_delay ~ origin + dest + hour + carrier + month + dow
mod_reg <- lm(flight_model, data = train)
tidy(mod_reg) %>% head(4)
```

```
## # A tibble: 4 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)   -16.0      6.12      -2.61 0.00905
## 2 originJFK       2.88     0.783      3.67 0.000239
## 3 destOAK        -4.61     3.10      -1.49 0.136
## 4 destSFO         1.89     0.620      3.05 0.00227
```

```
rmse <- function(y, yhat) sqrt(mean((y - yhat)^2))
rmse(test$arr_delay, predict(mod_reg, newdata = test))
```

```
## [1] 42.60868
```

`glmnet()` takes a design matrix; `alpha = 1` is the lasso, `0` ridge, values between the elastic net (`mixture` in tidymodels):

```
X_train <- model.matrix(flight_model, data = train)[, -1]
X_test <- model.matrix(flight_model, data = test)[, -1]
mod_lasso <- glmnet(X_train, train$arr_delay, alpha = 1)
coef(mod_lasso, s = 0.01)[1:4, ]
```

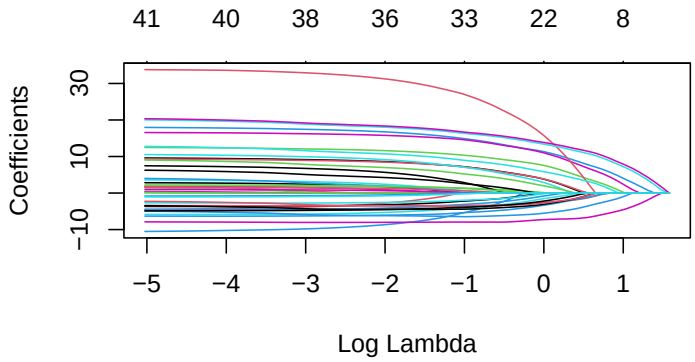
```
## (Intercept)   originJFK     destOAK     destSFO
##  -11.406671    2.783746   -4.399701    1.879023
```

```
rmse(test$arr_delay, predict(mod_lasso, newx = X_test, s = 0.01))
```

```
## [1] 42.60797
```

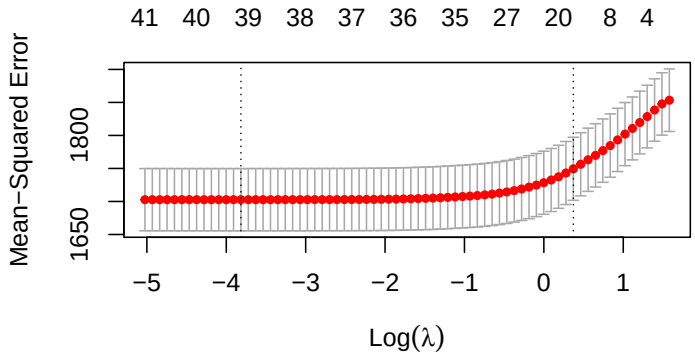
A small penalty barely moves the coefficients or test error: with 30 000 flights and 40 predictors least squares is not overfit. Coefficient paths:

```
plot(mod_lasso, xvar = "lambda")
```



10-fold CV (`cv.glmnet()`); the one-SE rule is sparser:

```
set.seed(1)
cv_lasso <- cv.glmnet(X_train, train$arr_delay, alpha = 1)
plot(cv_lasso)
```




```
c(lambda.min = cv_lasso$lambda.min, lambda.1se = cv_lasso$lambda.1se)
```

```
## lambda.min lambda.1se
##  0.0220299  1.4494202
```

```
sum(coef(cv_lasso, s = "lambda.1se") != 0)
```

```
## [1] 19
```

```
rmse(test$arr_delay, predict(cv_lasso, newx = X_test, s = "lambda.1se"))
```

```
## [1] 43.34445
```

Ridge keeps all predictors, shrunk:

```
mod_ridge <- glmnet(X_train, train$arr_delay, alpha = 0)
coef(mod_ridge, s = 0.1)[1:4, ]
```

```
## (Intercept)   originJFK     destOAK     destSFO
##   -7.683246    2.725897   -4.448581    1.861059
```

7.5 Selected exercises

Motivation. What survives of Proposition 7.1 for a general design, and where the C_p penalty comes from.

In Exercises 7.1 and 7.2, X is the $n \times p$ design matrix of centred predictors, y is centred (so no intercept is needed), and the objectives are written without the factor $1/n$, as in Proposition 7.1.

Exercise 7.1 (ridge in general)

Let $\hat\beta_\lambda$ minimise $\|y - X\beta\|^2 + \lambda\|\beta\|^2$ with $\lambda > 0$.
(a) Show that $\hat\beta_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ exists and is unique for every X, even when $p > n$.
(b) Show that $\|\hat\beta_\lambda\|$ is non-increasing in λ and tends to 0 as $\lambda \to \infty$.

Solution [Supplement].

(a) For $v \neq 0$, $v^\top(X^\top X + \lambda I)v = \|Xv\|^2 + \lambda\|v\|^2 > 0$, so $M = X^\top X + \lambda I$ is positive definite and invertible, whatever the rank of X. The objective equals $\beta^\top M\beta - 2\beta^\top X^\top y + \|y\|^2$, and completing

the square,

$$\beta^{\mathsf{T}} M\beta - 2\beta^{\mathsf{T}} X^{\mathsf{T}} y = (\beta - M^{-1}X^{\mathsf{T}}y)^{\mathsf{T}} M(\beta - M^{-1}X^{\mathsf{T}}y) - y^{\mathsf{T}} X M^{-1} X^{\mathsf{T}} y.$$

The first term is ≥ 0 with equality only at $\beta = M^{-1}X^{\mathsf{T}}y$, which is therefore the unique minimiser.

(b) Diagonalise $X^{\mathsf{T}}X = V\,\mathrm{diag}(d_1^2, \ldots, d_p^2)V^{\mathsf{T}}$ with V orthogonal and $d_j \geq 0$. Then $M^{-1} = V\,\mathrm{diag}\{1/(d_j^2+\lambda)\}V^{\mathsf{T}}$. With $z = V^{\mathsf{T}}X^{\mathsf{T}}y$, orthogonality of V gives

$$\|\hat\beta_\lambda\|^2 = \sum_{j=1}^{p} \frac{z_j^2}{(d_j^2+\lambda)^2},$$

each term non-increasing in λ and tending to 0. For $d_j = 1$ this is $1/(1+\lambda)$ (Proposition 7.1); small-d_j directions shrink most. ■

Exercise 7.2 (when the lasso selects nothing)

Let x_j be the jth column of X and $f(\beta) = \|y - X\beta\|^2 + \lambda\sum_j|\beta_j|$. Show that $\hat\beta = 0$ minimises f if and only if $\lambda \geq \lambda_{\max} = 2\max_j|x_j^{\mathsf{T}}y|$.

Solution [Supplement].

Step 1. (Sufficiency.) For any β,

$$\begin{aligned} f(\beta) - f(0) &= \|X\beta\|^2 - 2\sum_j \beta_j\, x_j^{\mathsf{T}} y + \lambda\sum_j|\beta_j| \\ &\geq \sum_j |\beta_j|(\lambda - 2|x_j^{\mathsf{T}}y|) \geq 0 \end{aligned}$$

when $\lambda \geq \lambda_{\max}$, using $\|X\beta\|^2 \geq 0$ and $\beta_j x_j^{\mathsf{T}}y \leq |\beta_j||x_j^{\mathsf{T}}y|$.

Step 2. (Necessity.) If $\lambda < 2|x_k^{\mathsf{T}}y|$ for some k, let $s = \mathrm{sgn}(x_k^{\mathsf{T}}y)$ and $\beta = ts\,e_k$ with $t > 0$. Then

$$f(\beta) - f(0) = t^2\|x_k\|^2 - t(2|x_k^{\mathsf{T}}y| - \lambda) < 0$$

for small t, so 0 is not a minimiser.

Step 3. With $X = I$ this is $|y_j| \leq \lambda/2$ (Proposition 7.1); `glmnet()` starts its path at $\lambda_{\max}$. ■

Exercise 7.3 (the optimism of the training error)

Let X have $q = d+1$ linearly independent columns (intercept included), $y = \mu + \epsilon$ with $\mu = X\beta$ and ϵ having uncorrelated components of mean 0 and variance σ^2, and let $y' = \mu + \epsilon'$ be new responses at the same x's, with ϵ' independent of ϵ and distributed like it. With $\hat y = Hy$, $H = X(X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}$, show that

$$\mathbb{E}\,\mathrm{RSS} = \mathbb{E}\|y - \hat y\|^2 = (n-q)\sigma^2, \qquad \mathbb{E}\|y' - \hat y\|^2 = (n+q)\sigma^2.$$

Solution [Supplement].

Step 1. H is symmetric and idempotent with $HX = X$, so $H\mu = \mu$; and $\mathrm{tr}H = q$, because $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ gives $\mathrm{tr}H = \mathrm{tr}\{(X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}X\} = \mathrm{tr}I_q$. For a fixed matrix B, $\mathbb{E}\,\epsilon^{\mathsf{T}}B\epsilon = \sum_{i,j} B_{ij}\mathbb{E}\epsilon_i\epsilon_j = \sigma^2\mathrm{tr}B$.

Step 2. $y - \hat y = (I-H)(\mu+\epsilon) = (I-H)\epsilon$, so $\mathrm{RSS} = \epsilon^{\mathsf{T}}(I-H)^{\mathsf{T}}(I-H)\epsilon = \epsilon^{\mathsf{T}}(I-H)\epsilon$ and $\mathbb{E}\,\mathrm{RSS} = \sigma^2\mathrm{tr}(I-H) = (n-q)\sigma^2$.

Step 3. $y' - \hat{y} = \epsilon' - H\epsilon$. Expanding, $\|y' - \hat{y}\|^2 = \|\epsilon'\|^2 - 2\epsilon'^{\mathsf{T}} H\epsilon + \epsilon^{\mathsf{T}} H\epsilon$. The cross term has mean 0 by independence and $\mathbb{E}\epsilon' = 0$, so the expectation is $n\sigma^2 + \sigma^2 \mathrm{tr} H = (n+q)\sigma^2$.

Step 4. The difference is $2q\sigma^2 = 2(d+1)\sigma^2$; C_p's $2d\hat{\sigma}^2$ corrects it up to the model-independent $2\sigma^2$. ■